

Start Hadoop

```
Start -dfs.sh  
Start -yarn.sh  
Jps
```

Start hive hive

1. Hive - Create Database

```
hive> SHOW DATABASES;  
OK  
default  
flightdb  
Time taken: 0.841 seconds, Fetched: 2 row(s)  
  
hive> CREATE DATABASE IF NOT EXISTS flightinfo;  
OK  
Time taken: 0.338 seconds  
hive> USE flightinfo;  
OK  
Time taken: 0.022 seconds
```

2. Hive - Drop Database

```
hive> drop database flightinfo cascade;  
OK  
Time taken: 156.316 seconds
```

A. Creating,Dropping,and altering Databasetables.

Internal Table

Create Table:

```
hive> CREATE TABLE employee (  
  >   empid SMALLINT,  
  >   ename STRING,  
  >   address STRING,  
  >   dob DATE  
  > )  
  > ROW FORMAT DELIMITED  
  > FIELDS TERMINATED BY ',';  
OK  
Time taken: -159.66 seconds
```

```
hive> DESCRIBE employee;  
OK  
empid          smallint
```

ename string
address string
dob date

Time taken: 160.786 seconds, Fetched: 4 row(s)

Hive - Alter Table

Rename a Table

```
hive> ALTER TABLE employee RENAME TO employee_details;
```

OK

Time taken: 0.228 seconds

Adding column

```
hive> ALTER TABLE employee_details ADD COLUMNS (salary INT);
```

OK

Time taken: 0.156 seconds

Change Column

```
hive> ALTER TABLE employee_details CHANGE ename emp_name STRING;
```

OK

Time taken: 0.156 seconds

Delete or Replace Column

```
hive> ALTER TABLE employee_details
```

```
> REPLACE COLUMNS (empid SMALLINT, ename STRING, address STRING);
```

OK

Time taken: 0.101 seconds

Dropping DatabaseTable

```
hive> DROP TABLE employee_details;
```

OK

Time taken: 0.592 seconds

B. Creating an external Hive table.

```
hadoop@DESKTOP-ND2BQ57:~$ hdfs dfs -mkdir /hivedirectory
```

```
hadoop@DESKTOP-ND2BQ57:~$ hdfs dfs -put ~/student.csv /hivedirectory
```

```
hadoop@DESKTOP-ND2BQ57:~$ hdfs dfs -ls /hivedirectory
```

Found 1 items

```
-rw-r--r--  1 hadoop supergroup    160 2026-04-22 08:45 /hivedirectory/student.csv
```

```
hadoop@DESKTOP-ND2BQ57:~$ hive
```

```
hive> CREATE EXTERNAL TABLE studentlist ( id INT, fname STRING, lname STRING,  
year STRING, major STRING )
```

```
> ROW FORMAT DELIMITED
```

```
> FIELDS TERMINATED BY ','
```

```
> LOCATION '/hivedirectory';
```

OK

Time taken: 0.372 seconds

```
hive> SELECT * FROM studentlist;
```

OK

```
1  jane  doe   senior mathematics
2  john  smith junior engineering
3  rahul patil senior it
4  priya sharma junior computer_science
5  amit  kumar sophomore    electronics
```

Time taken: 0.703 seconds, Fetched: 5 row(s)

Create the schema for a managed table.

```
hive> CREATE TABLE student ( id INT, fname STRING, lname STRING, year STRING,
major STRING ) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',';
```

OK

Time taken: 0.161 seconds

Move the external table data to the managed table.

```
hive> INSERT OVERWRITE TABLE student SELECT * FROM studentlist;
```

Query ID = hadoop_20260422084636_ecdef20e-1d78-47b9-8632-adbfc6981e7b

Total jobs = 3

Launching Job 1 out of 3

Total MapReduce CPU Time Spent: 0 msec

OK

Time taken: 2.343 seconds

```
hive> SELECT * FROM student;
```

OK

```
1  jane  doe   senior mathematics
2  john  smith junior engineering
3  rahul patil senior it
4  priya sharma junior computer_science
5  amit  kumar sophomore    electronics
```

Time taken: 0.343 seconds, Fetched: 5 row(s)

```
hive> DROP TABLE studentlist;
```

OK

Time taken: 0.458 seconds

C. Load table with data, insert new values and field in the table , Join tables with Hived.

```
hive> CREATE TABLE flightinfo2015 (YEAR SMALLINT, MONTH SMALLINT, DAY
SMALLINT, DAY_OF_WEEK SMALLINT, AIRLINE STRING, FLIGHT_NUMBER INT,
TAIL_NUMBER STRING, ORIGIN_AIRPORT STRING, DESTINATION_AIRPORT
STRING, SCHEDULED_DEPARTURE SMALLINT, DEPARTURE_TIME SMALLINT,
DEPARTURE_DELAY SMALLINT, TAXI_OUT SMALLINT, WHEELS_OFF
```

```
SMALLINT, SCHEDULED_TIME SMALLINT, ELAPSED_TIME SMALLINT,  
AIR_TIME SMALLINT, DISTANCE INT, WHEELS_ON SMALLINT, TAXI_IN  
SMALLINT, SCHEDULED_ARRIVAL SMALLINT, ARRIVAL_TIME SMALLINT,  
ARRIVAL_DELAY SMALLINT, DIVERTED SMALLINT, CANCELLED SMALLINT,  
CANCELLATION_REASON STRING, AIR_SYSTEM_DELAY SMALLINT,  
SECURITY_DELAY SMALLINT, AIRLINE_DELAY SMALLINT,  
LATE_AIRCRAFT_DELAY SMALLINT, WEATHER_DELAY SMALLINT ) ROW  
FORMAT DELIMITED FIELDS TERMINATED BY ',';
```

OK

Time taken: 0.132 seconds

```
hive> LOAD DATA LOCAL INPATH '/home/hadoop/flights.csv'  
> INTO TABLE flightinfo2015;
```

Loading data to table flightinfo.flightinfo2015

Table flightinfo.flightinfo2015 stats: [numFiles=1, totalSize=592406591]

OK

Time taken: 11.61 seconds

```
hive> CREATE TABLE airline ( iata_code STRING, airline STRING ) ROW FORMAT  
DELIMITED FIELDS TERMINATED BY ',';
```

OK

Time taken: 0.078 seconds

```
hive> CREATE TABLE airport ( iata_code STRING, airport STRING, city STRING, state  
STRING, country STRING, latitude STRING, longitude STRING ) ROW FORMAT  
DELIMITED FIELDS TERMINATED BY ',';
```

OK

Time taken: 0.061 seconds

```
hive> LOAD DATA LOCAL INPATH '/home/hadoop/airlines.csv' INTO TABLE airline;  
Loading data to table flightinfo.airline
```

Table flightinfo.airline stats: [numFiles=1, totalSize=359]

OK

Time taken: 0.662 seconds

```
hive> LOAD DATA LOCAL INPATH '/home/hadoop/airports.csv' INTO TABLE airport;  
Loading data to table flightinfo.airport
```

Table flightinfo.airport stats: [numFiles=2, totalSize=47734]

OK

Time taken: 0.332 seconds

Join table airline and flightinfo2015

```
hive> CREATE TABLE flight98 AS SELECT * FROM flightinfo2015 WHERE  
flight_number = 98;
```

Query ID = hadoop_20260422091315_e46b4383-d681-4b0e-b253-ce5ae41613ea

Total jobs = 3

Launching Job 1 out of 3
Total MapReduce CPU Time Spent: 0 msec
OK
Time taken: 11.068 seconds

JOIN Operation on Hive tables:

1. Display

year,month,day,day_of_week,airline,flight_number,origin_airport,destination_airport, airline name by joining flight98 with airline.(Inner join).

```
hive> SELECT f.year, f.month, f.day, f.day_of_week, f.airline, f.flight_number,  
f.origin_airport, f.destination_airport, a.airline FROM flight98 f JOIN airline a ON f.airline =  
a.iata_code;
```

Query ID = hadoop_20260422124554_ad5247da-e643-46dc-bc07-52da355463c2

Total jobs = 1

MapReduce Jobs Launched:

Stage-Stage-3: HDFS Read: 1185334975 HDFS Write: 489025 SUCCESS

Total MapReduce CPU Time Spent: 0 msec

OK

```
hive> SELECT * FROM flight98 f LEFT JOIN airline a ON f.airline = a.iata_code;
```

OK

Time taken: 0.132 seconds

2. Display information about the airline which will not provide services by joining flight98 & airline. (Right Outer Join).

```
hive> SELECT * FROM flight98 f RIGHT JOIN airline a ON f.airline = a.iata_code  
WHERE f.flight_number IS NULL;
```

OK

Time taken: 169.345 seconds, Fetched: 22 row(s)

3.List all record by joining flight98 and airline.(full outer join).

```
hive> SELECT * FROM flight98 f FULL OUTER JOIN airline a ON f.airline = a.iata_code;  
SELECT * FROM flight98 f FULL OUTER JOIN airline a ON f.airline = a.iata_code;
```

OK

Time taken: 2.667 seconds, Fetched: 4554 row(s)

4.Create index on origin_airport field

```
hive> CREATE INDEX f08_index ON TABLE flightinfo2015 (Origin_airport) AS  
'COMPACT' WITH DEFERRED REBUILD;
```

```
hive> CREATE INDEX origin_index ON TABLE flightinfo2015 (origin_airport) AS  
'COMPACT' WITH DEFERRED REBUILD;
```

OK

Time taken: 0.239 seconds

5. Find average departure delay for flight number 98 day of week wise.

```
hive> SELECT day_of_week, AVG(departure_delay) FROM flightinfo2015 GROUP BY day_of_week;
```

Total MapReduce CPU Time Spent: 0 msec

OK

NULL NULL

5 9.429767977851792

2 9.164522265904088

7 9.402855627605891

1 10.871775063646947

6 7.827415699173568

3 8.645276690756555

4 9.956913497060805

Time taken: 25.712 seconds, Fetched: 8 row(s)

```
hive> SELECT day_of_week, AVG(departure_delay) FROM flightinfo2015 WHERE flight_number = 98 GROUP BY day_of_week;
```

Total MapReduce CPU Time Spent: 0 msec

OK

5 11.35

2 5.21301775147929

7 13.589887640449438

1 12.397660818713451

6 10.728813559322035

3 12.616438356164384

4 9.527777777777779

Time taken: 12.149 seconds, Fetched: 7 row(s)

Employee queries

1. CREATE DATABASE

```
hive> CREATE DATABASE IF NOT EXISTS mydb;CREATE DATABASE IF NOT EXISTS mydb;
```

OK

Time taken: 4.971 seconds

OK

Time taken: 0.087 seconds

```
hive> SHOW DATABASES;
```

OK

default

flightdb

flightinfo

mydb

Time taken: 1.005 seconds, Fetched: 4 row(s)

2. DROP DATABASE

```
hive> DROP DATABASE IF EXISTS mydb;
OK
Time taken: 0.697 seconds
hive> DROP DATABASE IF EXISTS userdb CASCADE;
OK
Time taken: 0.031 seconds
hive> CREATE DATABASE IF NOT EXISTS userdb;
OK
Time taken: 160.619 seconds
hive> DROP SCHEMA IF EXISTS userdb;
OK
Time taken: 0.067 seconds
```

3. Create Table

```
hive> CREATE TABLE IF NOT EXISTS employee ( eid INT, name STRING, salary
FLOAT, destination STRING ) COMMENT 'Employee details' ROW FORMAT
DELIMITED FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' STORED AS
TEXTFILE;
OK
Time taken: -157.643 seconds
hive> SHOW TABLES;
OK
employee
flightinfo2007
Time taken: 0.06 seconds, Fetched: 2 row(s)
```

4. Alter Table

```
hive> ALTER TABLE employee RENAME TO emp;
OK
Time taken: -159.497 seconds
hive> ALTER TABLE emp CHANGE name ename STRING;
OK
Time taken: 0.237 seconds
hive> ALTER TABLE emp CHANGE salary salary DOUBLE;
OK
Time taken: 0.137 seconds
hive> DESCRIBE emp;
OK
eid                int
ename              string
salary             double
destination         string
Time taken: 0.416 seconds, Fetched: 4 row(s)
```

```
hive> ALTER TABLE emp ADD COLUMNS (age INT);
OK
Time taken: 0.155 seconds
hive> ALTER TABLE emp REPLACE COLUMNS ( eid INT, ename STRING, salary
DOUBLE );
OK
Time taken: 0.139 seconds
hive> SHOW TABLES;
OK
emp
flightinfo2007
Time taken: -160.401 seconds, Fetched: 2 row(s)
hive> DESCRIBE emp;
OK
eid                int
ename              string
salary             double
Time taken: 0.078 seconds, Fetched: 3 row(s)
```

5. Add column statement

```
hive> ALTER TABLE emp ADD COLUMNS ( dept STRING COMMENT 'Department
name' );
OK
Time taken: 0.182 seconds
hive> DESCRIBE emp;
OK
eid                int
ename              string
salary             double
dept              string      Department name
Time taken: 0.115 seconds, Fetched: 4 row(s)
```

6. REPLACE STATEMENT

```
hive> ALTER TABLE emp REPLACE COLUMNS ( empid INT, name STRING );
OK
Time taken: 0.144 seconds
hive> DESCRIBE emp;
OK
empid              int
name               string
Time taken: 0.104 seconds, Fetched: 2 row(s)
```

7. DROP TABLE

hive> DROP TABLE IF EXISTS emp;

OK

Time taken: 0.995 seconds

hive> SHOW TABLES;

OK

flightinfo2007

Time taken: 0.061 seconds, Fetched: 1 row(s)

8. Index

hive> CREATE TABLE emp (eid INT, name STRING, salary FLOAT, destination STRING) ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t' STORED AS TEXTFILE;

OK

Time taken: 0.631 seconds

hive> ALTER TABLE emp ADD COLUMNS (yoy INT);

OK

Time taken: 0.11 seconds

hive> CREATE INDEX index_yoy ON TABLE emp (yoy) AS

'org.apache.hadoop.hive.ql.index.compact.CompactIndexHandler' WITH DEFERRED REBUILD;

OK

Time taken: 0.302 seconds

hive> ALTER INDEX index_yoy ON emp REBUILD;

Query ID = hadoop_20260428115126_ad755389-a7eb-47ed-bfbe-82b1274226f8

Total jobs = 1

Launching Job 1 out of 1

Total MapReduce CPU Time Spent: 0 msec

OK

Time taken: 9.128 seconds

hive> SHOW INDEX ON emp;

OK

index_yoy	emp	yoy	default__emp_index_yoy__compact
-----------	-----	-----	---------------------------------

Time taken: 0.051 seconds, Fetched: 1 row(s)

9. DROP INDEX

hive> DROP INDEX index_yoy ON emp;

OK

Time taken: 0.159 seconds

hive> SHOW INDEX ON emp;

OK

Time taken: 0.042 seconds

hive>

10. Select ... order by

hive> INSERT INTO emp VALUES

> (1, 'Amit', 50000, 'Manager', 'IT'),
> (2, 'Neha', 30000, 'HR', 'HR'),
> (3, 'Rahul', 40000, 'Developer', 'IT'),
> (4, 'Sneha', 35000, 'Tester', 'QA');

Query ID = hadoop_20260428115700_4361d4fa-5de4-4fe7-80db-af1c96680034

Total jobs = 3

Launching Job 1 out of 3

Total MapReduce CPU Time Spent: 0 msec

OK

Time taken: 2.733 seconds

hive> SELECT * FROM emp ORDER BY salary;

Query ID = hadoop_20260428115714_c927dabf-2436-4115-b1b0-a613d6995190

Total jobs = 1

Launching Job 1 out of 1

Total MapReduce CPU Time Spent: 0 msec

OK

2	Neha	30000.0	HR	NULL
4	Sneha	35000.0	Tester	NULL
3	Rahul	40000.0	Developer	NULL
1	Amit	50000.0	Manager	NULL

Time taken: 2.45 seconds, Fetched: 4 row(s)

hive> SELECT * FROM emp ORDER BY salary DESC;

Total MapReduce CPU Time Spent: 0 msec

OK

1	Amit	50000.0	Manager	NULL
3	Rahul	40000.0	Developer	NULL
4	Sneha	35000.0	Tester	NULL
2	Neha	30000.0	HR	NULL

Time taken: 1.434 seconds, Fetched: 4 row(s)

11. Select... group by

hive> SELECT dept, COUNT(*) FROM emp GROUP BY dept;

Total MapReduce CPU Time Spent: 0 msec

OK

HR 1

IT 2

QA 1

Time taken: 2.975 seconds, Fetched: 3 row(s)

```
hive> SELECT dept, SUM(salary) FROM emp GROUP BY dept;
```

```
Total MapReduce CPU Time Spent: 0 msec
```

```
OK
```

```
HR    30000.0
```

```
IT    90000.0
```

```
QA    35000.0
```

```
Time taken: 1.676 seconds, Fetched: 3 row(s)
```

```
hive> SELECT dept, COUNT(*) FROM emp GROUP BY dept HAVING COUNT(*) > 1;
```

```
Total MapReduce CPU Time Spent: 0 msec
```

```
OK
```

```
IT    2
```

```
Time taken: 1.409 seconds, Fetched: 1 row(s)
```

```
hive> SELECT dept, AVG(salary) FROM emp GROUP BY dept;
```

```
Total MapReduce CPU Time Spent: 0 msec
```

```
OK
```

```
HR    30000.0
```

```
IT    45000.0
```

```
QA    35000.0
```

```
Time taken: 1.297 seconds, Fetched: 3 row(s)
```

```
hive> SELECT * FROM emp ORDER BY salary DESC;
```

```
Total MapReduce CPU Time Spent: 0 msec
```

```
OK
```

```
1    Amit    50000.0 Manager IT    2020
```

```
3    Rahul    40000.0 Developer IT    2019
```

```
4    Sneha    35000.0 Tester QA    2022
```

```
2    Neha    30000.0 HR    HR    2021
```

```
Time taken: 161.799 seconds, Fetched: 4 row(s)
```

11. JOINS

```
hive> CREATE TABLE customer ( id INT, name STRING, age INT, address STRING,  
salary INT ) ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t';
```

```
OK
```

```
Time taken: 0.287 seconds
```

```
hive> CREATE TABLE orders ( oid INT, order_date STRING, c_id INT, amount INT )  
ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t';
```

```
OK
```

```
Time taken: 0.119 seconds
```

```
hive> INSERT INTO customer VALUES
```

```
> (1, 'Kavita', 24, 'Sangvi', 34000),
```

```
> (2, 'Chatur', 23, 'Kothrud', 35000),  
> (3, 'Fatema', 31, 'Lohgad', 20000),  
> (4, 'Rohan', 27, 'Pune Station', 22000);
```

Total MapReduce CPU Time Spent: 0 msec

OK

Time taken: 2.725 seconds

```
hive> SELECT c.id, c.name, c.age, o.amount FROM customer c JOIN orders o ON (c.id =  
o.c_id);
```

Total MapReduce CPU Time Spent: 0 msec

OK

```
2    Chatur 23    3420  
3    Fatema 31    1200  
3    Fatema 31    3400  
4    Rohan  27    2150
```

Time taken: 171.684 seconds, Fetched: 4 row(s)

12. LEFT OUTER JOIN

```
hive> SELECT c.id, c.name, o.amount FROM customer c LEFT OUTER JOIN orders o ON  
(c.id = o.c_id);
```

Total MapReduce CPU Time Spent: 0 msec

OK

```
1    Kavita NULL  
2    Chatur 3420  
3    Fatema 1200  
3    Fatema 3400  
4    Rohan  2150
```

Time taken: 15.855 seconds, Fetched: 5 row(s)

13. RIGHT OUTER JOIN

```
hive> SELECT c.id, c.name, o.amount FROM customer c RIGHT OUTER JOIN orders o  
ON (c.id = o.c_id);
```

Total MapReduce CPU Time Spent: 0 msec

OK

```
3    Fatema 1200  
3    Fatema 3400  
4    Rohan  2150  
2    Chatur 3420
```

Time taken: 8.244 seconds, Fetched: 4 row(s)

14. WHERE Clause (Filtering)

```
hive> SELECT * FROM emp WHERE salary > 35000;
```

OK

```
1    Amit    50000.0 Manager IT    2020
```

3 Rahul 40000.0 Developer IT 2019
Time taken: 2.671 seconds, Fetched: 2 row(s)

15. DISTINCT (Unique values)

hive> SELECT DISTINCT dept FROM emp;
Total MapReduce CPU Time Spent: 0 msec
OK
HR
IT
QA
Time taken: 4.064 seconds, Fetched: 3 row(s)

16. Aggregate Functions

hive> SELECT MAX(salary) FROM emp;
Total MapReduce CPU Time Spent: 0 msec
OK
50000.0
Time taken: 162.482 seconds, Fetched: 1 row(s)

hive> SELECT MIN(salary) FROM emp;
Total MapReduce CPU Time Spent: 0 msec
OK
30000.0
Time taken: 1.467 seconds, Fetched: 1 row(s)

hive> SELECT AVG(salary) FROM emp;
Total MapReduce CPU Time Spent: 0 msec
OK
38750.0
Time taken: 1.405 seconds, Fetched: 1 row(s)

hive> SELECT SUM(salary) FROM emp;
Total MapReduce CPU Time Spent: 0 msec
OK
155000.0
Time taken: 1.369 seconds, Fetched: 1 row(s)

17. JOIN with WHERE

hive> SELECT c.name, o.amount FROM customer c JOIN orders o ON c.id = o.c_id
WHERE o.amount > 2000;
Total MapReduce CPU Time Spent: 0 msec
OK
Chatur 3420
Fatema 3400
Rohan 2150
Time taken: 11.267 seconds, Fetched: 3 row(s)

18. ORDER BY Multiple Columns

```
hive> SELECT * FROM emp ORDER BY dept ASC, salary DESC;
```

Total MapReduce CPU Time Spent: 0 msec

OK

2	Neha	30000.0	HR	HR	2021
1	Amit	50000.0	Manager	IT	2020
3	Rahul	40000.0	Developer	IT	2019
4	Sneha	35000.0	Tester	QA	2022

Time taken: 162.252 seconds, Fetched: 4 row(s)